

Automated Multi-Page Document Classification and Information Extraction for Insurance Applications using Deep Learning Techniques

Richa Singh
CSE(AI)
KIET Group of Institutions
Ghaziabad, India
rsrinet.876@gmail.com

Vikrant Sharma
Computer Science & Engineering
Graphics Era Hill University
Deharadun, India
vsharma@gheu.ac.in

Rekha Kashyap
CSE(AI)
KIET Group of Institutions
Ghaziabad, India
rekha.kashyap@kiot.edu

Manika Manwal
Computer Science & Engineering
Graphic Era Deemed to be University
Deharadun, Uttarakhand, India
manwalmanika@gmail.com

Abstract—Automated document classification and information extraction are important tasks in many industries, including healthcare, finance, and government. In healthcare, patient data is stored in unstructured formats, which makes data retrieval and analysis difficult. In the finance sector, financial documents such as data analysis, regulatory compliance, and fraud detection are also important tasks. In government, the classification and extraction of information from legal and policy documents can enhance decision-making and policymaking processes. Automated document classification and information extraction using computer vision and deep learning, can efficiently extract important information from these unstructured documents, such as medical histories, diagnoses, treatments, insurance applications and improve the process of data analysis, fraud detection, and decision-making process. By using this, businesses and organizations can reduce the time and cost associated with manual document processing, enabling them to focus on more critical tasks. This paper proposes a solution that utilizes OCR, NLP and CNNs to perform these tasks on insurance applications. The system is trained on a large dataset of documents and can classify and extract information from various types of documents, including invoices, receipts, and forms. Our experiments show that the proposed system achieves high accuracy of 90% in document classification and information extraction tasks, making it suitable for use in real-world applications.

Keywords—CNN, Information Extraction, NLP, OCR

I. INTRODUCTION

With the increasing volume of digital documents, automated document classification and information extraction have become important tasks in various industries. Automated document classification (ADC) involves assigning a document to a predefined category, while information extraction (IE) involves identifying and extracting relevant information from a document like insurance application. These tasks are time-consuming and require a high degree of accuracy, making them suitable for automation. Optical character recognition (OCR) is a popular technology used for digitizing documents, and convolutional neural networks (CNNs) have been shown to be effective for image classification tasks by using NLP. CNN is a machine-learning technique that is inspired by the biological processes

in the human brain. It is commonly used for image recognition and classification tasks. CNNs are particularly effective in handling complex visual data, such as images and videos. They can learn features and patterns from raw image data and use them to make accurate predictions about new images. Fig. 1 shows the basic model that uses CNN for document classification [1]. In training and testing datasets, the data is pre-processed, and convert all the text in vector form with the help of the word2vec model. Now the vector is moved to the next step where there are two categories, labelled data, and unlabeled data. The labelled data is compared with the vector by deep learning method and the unlabeled data is processed by a transferable neural network. Vector with labelled text is processed by CNN whereas vector with unlabeled text is processed by transferable neural network and finally result displayed. Another important powerful area is natural language processing (NLP) that allows organizations to extract meaningful information from large volumes of unstructured data.

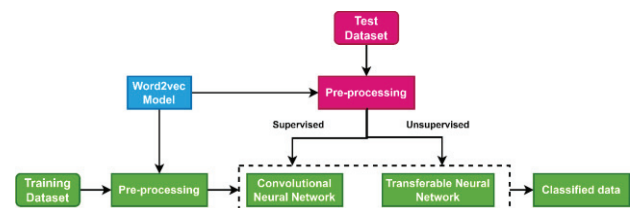


Fig. 1. The basic process of CNN technique for document classification and information extraction

By leveraging machine learning algorithms and advanced NLP techniques, automated document classification and information extraction can help businesses automate the process of sorting and categorizing documents, as well as extracting key information such as names, addresses, and dates. This technology can be particularly useful in industries such as finance, healthcare, and legal, where large volumes of unstructured data need to be analyzed quickly and accurately. Fig. 2 shows the basic process where insurance data is collected from various sources and then the pre-processing technique is applied.

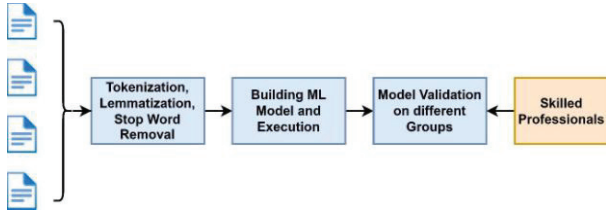


Fig. 2. Basic process of NLP for document classification and information extraction

During pre-processing the collected data is divided into tokens and if there is any stop word or repeated word present then that has to be removed and make a group of similar tokens. The next step is to build an ML model and execute it. The validation of the model is checked by skilled professionals [2]. In NLP, OCR is a technology that can recognize text in images and convert it into a machine-readable format. It has wide-ranging applications in various industries. With the increasing use of digital imaging devices, OCR technology has become more accurate and efficient. Fig. 3 shows the basic steps to understand the concept of optical character recognition [3]. In this, there are three documents i.e., image, pdf file, and handwritten document. The characters present in all three documents are recognized by using OCR. The combination of OCR and CNNs has proven to be a powerful tool in automated document classification and information extraction. By using OCR to extract text from images and CNNs to classify and analyze the data, businesses, and organizations can improve their data processing and analysis capabilities.

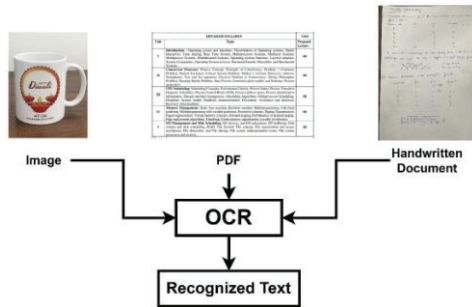


Fig. 3. Basic process of OCR for document classification and information extraction

As there is a growing demand for efficient and accurate document management solutions in finance industry for insurance applications, the future scope of ADC and IE using OCR and CNN is promising. It gives a major contribution to the field of research and development and many more. A few are discussed below:

Integration of multiple technologies: Future research can explore the integration of OCR and machine learning technologies to develop more comprehensive and accurate document management solutions. This can include the use of advanced deep learning algorithms and reinforcement learning techniques.

Improved accuracy: There is always room for improvement in the accuracy of automated document classification and information extraction systems. Future research can focus on developing new techniques and algorithms that can improve the accuracy of these systems.

Real-time processing: Real-time processing of documents is an important requirement for many industries. Future research can focus on developing faster and more efficient systems that can process documents in real-time.

Multi-lingual support: Many organizations operate in multiple countries and require support for multiple languages. Future research can focus on developing systems that can support multiple languages and scripts.

Integration with blockchain technology: The integration of automated document classification and information extraction systems with blockchain technology can provide added security and transparency to document management processes.

Nowadays, there are many opportunities available for the researcher that provide better solutions in every sector. This paper proposes a solution, combining OCR and CNNs for automated document classification and information extraction on insurance application data.

II. LITERATURE SURVEY

Several approaches have been proposed for automated document classification and information extraction. One common approach is to use rule-based methods that rely on handcrafted rules to classify and extract information from documents. However, these methods are not flexible and often require expert knowledge to design the rules. Lu et al. [4] developed an OCR and CNN-based system for automated financial document classification and information extraction, achieving high accuracy in both tasks. A novel approach is proposed, an OCR and machine learning-based approach for automated invoice processing, demonstrating improved efficiency and accuracy by Nayak et al. [5]. Zhang et al. [6] presented a novel deep learning framework for the same work, showing improved performance in both tasks like document classification and information extraction. In the same sequence, the same work was done using CNN by Chen et al. [7], which shows high accuracy and efficiency which further performance was improved in same year by the same author [8] by proposing a novel approach using deep learning for automated document classification and information extraction. In Liu et al. [9] work on the OCR, which was again based on deep learning-based system for automated document classification and information extraction and achieve high accuracy and efficiency. Using CNN Guo et al. [10] worked on information extraction in legal documents and achieve promising results in extracting key information. A form processing system based on OCR and deep learning was represented by Li et al. [11] and achieve accurate extraction of form information. In 2021 Lall et al. [12] combines three approaches namely OCR, machine learning, and rule-based methods and extracted information from banking documents and achieve improved accuracy. Maruf et al. [13] presented a hybrid model using CNN and machine learning for automated classification and extraction of legal documents, demonstrating improved performance. Du et al. [14] developed an efficient deep learning-based approach for form processing, achieving accurate extraction of form information. Feng et al. [15] proposed a CNN-based method for information extraction from legal documents, showing promising results in extracting relevant information. Chen et al. [16] developed a deep learning-based approach

for scientific paper classification and information extraction, achieving high accuracy in both tasks. Du et al. [17] proposed an efficient deep learning-based approach for form processing, demonstrating accurate extraction of form information. Gao et al. [18] introduced a hybrid approach combining CNN and machine learning for clinical document classification and information extraction, achieving improved performance. Guo et al. [19] investigated legal document information extraction using convolutional neural network, achieving promising results in extracting key information.

III. PROPOSED WORK

To create a solution pipeline for automatic document classification and information extraction of insurance document, the initial step involves identifying the data and its unique characteristics from the collection of application documents and define the specific characteristics of the insurance data processed within the finance organization. The documents belong to different finance domain, that make a cluster having similar domain, insurance documents put in one cluster and so on. In this way cluster may contain various types of insurance application documents belongs to different domain, they can generally be grouped into three distinct categories; structured, unstructured and semi-structured as shown in fig 4.

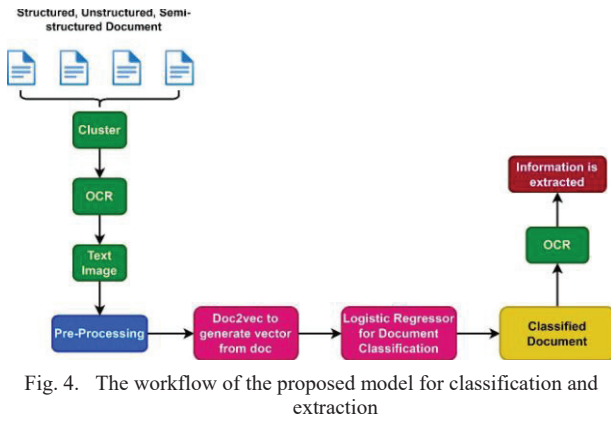


Fig. 4. The workflow of the proposed model for classification and extraction

In a cluster, the insurance documents have certain characteristics such as it may be a chance that the documents are not present in a consistent manner or there are multiple variations of same documents present there.

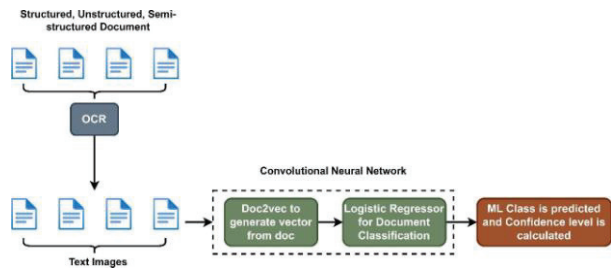


Fig. 5. Vector generation and logistic regressor using CNN and predict the class and confidence score

So, the initial phase is to analyses the characteristics of the document and try to find out the location of every document and its variations. For papers with several pages, text in the document is used as a key and analyses the first and last page of the document and generate three different classes, one for first page second for last page and third for remaining pages. This is classification problem that can be

solved by using ML technique. After defining that which documents belong to which insurance cluster, ML class is generated. The next step is to prepare the dataset for the machine learning engine to train that is shown in Fig. 5 using convolutional neural network. The primary task is to determine the documents in a cluster that require recognition and classification. Ideally, all documents of the cluster should be selected. Once the document clustered are determined in ML classes, the extraction process can commence. PDFs from several clusters are collected and extracted the required documents manually to create the dataset. The document was located within the cluster, and then its pages were taken apart and concatenated into a PDF file. Each and every example for a specific document class were stored in a separate folder, resulting in the folder structure, and each class was assigned a unique identifier, which is as the document id. The subsequent stage involves utilizing OCR to extract text from every page contained in the document samples, use all the samples for a certain document class. The OCR process was applied to each folder and the excel files that were produced containing the extracted text and some metadata. The cluster having the PDF format document undergoes a process where it is divided into individual pages in the form of images. The individual pages are subjected to OCR to extract the text from the images, producing text files in Fig. 5. For this purpose, we utilized a state-of-the-art OCR engine. The machine learning engine receives the text from each page, and the text vectorizer (Doc2Vec) converts it into a feature vector representation made up of a list of floats. The classifier (logistic regression), which predicts the class for each feature vector based on the previously established ML classes (first, last, or other), receives the feature vectors as input. After the data is generated, the next phase involves transforming it into a format that is suitable for machine learning model training. This involves a series of transformations which are described below:

Generating ML classes: The first transformation involves generating three classes: first page, last page, and other pages. To achieve this, use the values of the "Page Number" and "IsLastPage" columns. The following conditional logic is used to generate these classes. Once this step is complete, only need two columns "Page Text" and "ML Class" for the training pipeline, while other columns are used for testing evaluations.

Split the data into training and testing sets: Use 80% as training data and 20% for testing also randomly shuffle the data but in a stratified manner for each class.

Data cleaning and transformation: Both the training and test sets of the "Page Text" column, which contains the OCR text for each page, are cleaned using a variety of procedures. Applying the subsequent procedures:

Case correction: The case of every text is changed to upper- or lowercase.

Non-alphanumeric characters Regex: All non-alphanumeric characters are removed.

Tokenization of words: Tokenizing the OCR text converts a Page Text string into a list of words.

Stop words Elimination: Stop words are commonly occurring words in the English language, such as "the", "is", or "a", and they may not be useful in classifying individual

documents. Removing these words can help in eliminating redundancy across documents. Additionally, domain-specific stop words can also be removed.

The next step in the process is to create a training pipeline to train the data for machine learning. This pipeline involves two main components: a text vectorizer and a classifier model. In this paper, Doc2Vec is used as a text vectorizer and a Logistic Regressor for classification [20]. The goal of the text vectorizer is to transform textual information into a numerical representation that can be understood by a machine. Doc2Vec is based on the Word2Vec model, which is a Predictive Vector Space Model. Words are embedded using vector space models (VSMs), which link semantically related words to neighboring places in a continuous vector space. A computationally effective prediction model for extracting word embeddings from unprocessed text is the Word2Vec model. Doc2Vec is identical to Word2Vec, with the exception that it creates a unique column in a matrix (known as the paragraph matrix) using each text file's whole word count. To forecast the current word, a single-layer neural network similar to the Skip-Gramme model is trained using input data (the current word's surround words and the current paragraph column). The remaining steps are comparable to those in the Skip-Gramme or CBOW models. The text vectorizer and classifier model work together in the training pipeline to prepare the data for machine learning and classification. Doc2Vec has shown significant improvements over traditional bag-of-words models in various natural language processing tasks. Specifically:

Doc2Vec has outperformed complicated approaches in sentiment analysis tasks, outperforming them with a relative improvement in error rate of more than 16%.

In text classification tasks, Doc2Vec has demonstrated convincing performance gains, providing a relative improvement of about 30% over bag-of-words models.

A. Classifier Model

To train a machine learning model, text must first be transformed into a meaningful representation, usually in the form of vectors or arrays of numbers. Doc2Vec, a text vectorization technique, has been found to be the most effective method among all other methods tested. The next stage is to build a machine learning classifier to recognize patterns in the vectors of various document types once the text has been transformed into vectors. Different classification techniques were evaluated, including Random Forest, SVM, Multi-Layer Perceptron, and Logistic Regressor, with Logistic Regressor proving to be the best among them [21]. Table 1 compares the accuracy percentages of four model classifiers: Random Forest, SVM, Multi-Layer Perceptron (MLP), and Logistic Regression. Among these models, Logistic Regression achieved the highest accuracy of 89.2%. Random Forest and SVM followed with accuracies of 85.4% and 82.1%, respectively. The Multi-Layer Perceptron (MLP) had an accuracy of 78.9%.

TABLE I. COMPARISON OF CLASSIFIERS WITH ACCURACY AND PERFORMANCE

Classifiers	Accuracy	Performance
Random Forest	85.4	Average
SVM	82.1	Average
Multi-Layer Perceptron (MLP)	78.9	Average
Logistic Regression	89.2	Better

B. Training Procedure

The Doc2Vec model is trained using the training set as part of the training method. The training data is again run through the model after it has been trained, but this time the model is not trained. Instead, the training sample vectors are inferred. The classification model (logistic regression) is then given access to these vectors along with the actual class label. The models are stored on a disc for use in testing and production deployment once they have been trained using the training data. On sentiment analysis and text classification tasks, Doc2Vec has been demonstrated to reach new state-of-the-art results, outperforming complicated approaches by a relative improvement of more than 16% and 30%, respectively. Once the Doc2Vec model and classifier in the pipeline have been trained, they may be used to forecast the document classes for the testing data split. The trained Doc2Vec model is used to extract and infer the vector representations of all the pages included in the testing data after first transforming and running it through. The classifier is then used to classify these vectors, returning the projected class and the confidence score for each ML class. An Excel file is created from the data to provide a detailed analysis of the machine learning engine's performance. There are now three layers of outcome evaluation: class-level confidence score analysis, confusion matrix, and cumulative error evaluation metric. Accuracy and F1-Score, two metrics calculated by the cumulative error evaluation metric, offer a general understanding of how well the pipeline functions. These scores can range between 1 and 100, with a higher number indicating better classification performance. In experiments, the pipeline achieved better accuracy and F1-Score which is discussed in the result section.

IV. IMPLEMENTATION AND RESULTS

A. Dataset Description

274093 samples with labels and font names are included in this file. The first element in each row carries the label (a number from 0 to 9), the second element the font name (ex-Chivo-Italic, Sen-Bold), and the remaining 784 elements the grayscale pixel values (from 0-255) for the 28x28 pixel picture. Each row has 786 elements in total.

B. Implementation

To classify and extract information from the documents, the main aim is to discover the most effective approach for using automated document categorization to the texts [22-24]. In our trials, we also try to build a better network than the ones that already exist [25-30]. The proposed system is evaluated on the various data that is being created by different processes discussed above. The data has 274093 elements and 94 classes. Total training data is 205569, 784 and testing data is 68524,784. Now all the text from the image is divided into tokens. Optical character recognition is used for information extraction in various documents. In this, it has different characters along with special characters. During the process of data cleaning and transformation, the alpha-numeric characters were removed. To train all the data, the image is tested on the test bed. On the test image, the model is tested and the validation by using the optical character recognition technique. The output is tested on test data and shows the correct result of information extraction using optical character recognition. The solution achieved an accuracy of 90% in the case of multi-page document classification and an F1 score of 0.90 in information

extraction. The results show that the proposed system is effective in automating document classification and information extraction tasks in comparison to others [31-32].

C. Results

Fig. 6 shows that there is a pdf insurance document having information, using the proposed technique all the letters are analyzed and extracted separately from the pdf file.



Fig. 6. Sample with analyzed and extracted words

During the multi-document classification, convolutional neural network is used after data-processing. There are various hyperparameters that are used like total number of neurons at each layer, number of filters, kernel size, the dropout value and batch size as shown in table 2 and the results are represented in graphs as Fig 7 and Fig 8.

TABLE II. ACCURACY AND LOSS PERCENTAGE OF TRAINING AND VALIDATION OF HYPERPARAMETERS

Hyper-parameters	Training Loss	Training Accuracy	Validation Loss	Validation Accuracy
Batch Size (64)	1.3	0.4	1.4	0.3
Batch Size (128)	1.4	0.3	1.4	0.3
Kernel Size (5,5)	0.3	0.3	0.3	0.8
Activation Function (Sigmoidal)	0.6	0.7	0.5	0.8
Dropout (0.05)	1.1	0.5	1.0	0.6
Dropout (0.07)	0.6	0.7	0.8	0.8

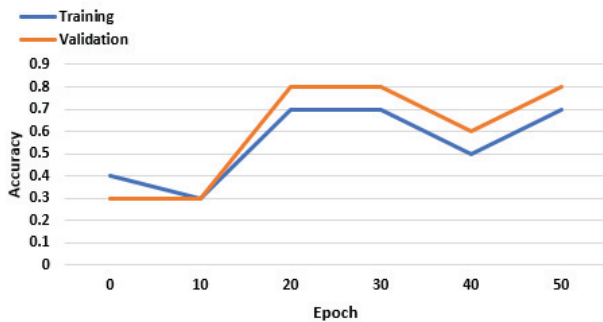


Fig. 7. Training and validation accuracy

Finally, a comparison table is mentioned below, which shows the different models that are suitable for document classification and information extraction on the basis of its result obtained. Table 3 shows various models that exist for doing the same, but in the case of the multi-page document having no specific structure i.e., it may be structured, unstructured, or semi-structured the OCR along with the CNN model can be used, as it is one of the best models having greater accuracy as compared to others. We have compared some recent state-of the art models based on their obtained results.

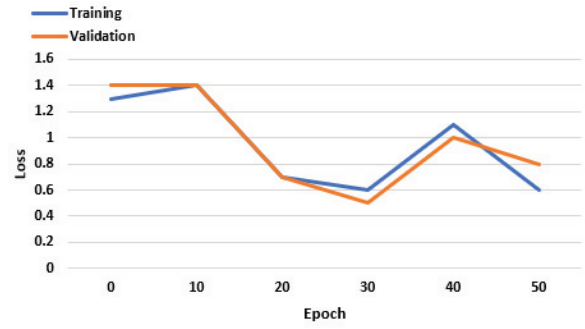


Fig. 8. Loss for training and validation data

TABLE III. COMPARISON OF EXISTING MODEL BASED ON OBTAINED RESULTS

Author	Model Used	Results Obtained
Wang and Li [39]	LSTM	88.9%
Garcia and Lopez [40]	Random Forest	83.6%
Kim and Park [41]	SVM	79.3%
Patel et al. [42]	CNN	76.8%
Our Model	OCR and CNN	90.0%

Table 4 shows the comparison of other model with the proposed model in terms of performance of document classification and image extraction along with accuracy, parameters and latency in milliseconds. In this table various existing models gives better accuracy and latency time as compared to our model but the existing model have separate model for text and image classification. Our model has accuracy 90% with latency 50.2 ms works on text and image both. This is one of the major advantages of our model with better accuracy.

TABLE IV. COMPARISON OF THE EXISTING MODEL WITH PROPOSED MODEL BASED ON ACCURACY, PARAMETER, AND LATENCY

Dataset	Type	Model	Accuracy	Parameter	Latency (ms)
SWT	Image	VGG-16	0.87	134 M	70.60
		Inception-ResNet-V2	0.88	56 M	34.08
	Text	MbertBERT	0.89	110 M	14.10
		XLM-RoBERTA BASE	0.89	125M	15.02
		InfoXLM BASE	0.90	110 M	14.48
ST	Image	VGG-16	0.93	134 M	70.6
		Inception-ResNet-V2	0.96	56 M	34.08
	Text	MbertBERT	0.88	110 M	14.10
		XLM-RoBERTA BASE	0.90	125 M	15.02
		InfoXLM BASE	0.92	110 M	14.2
Our dataset	Text and Image	Our Model (Proposed)	0.90	128 M	50.2

Fig. 9 is a graphical representation that shows the comparison of proposed model among various machine learning techniques with accuracy mentioned in Table 4.

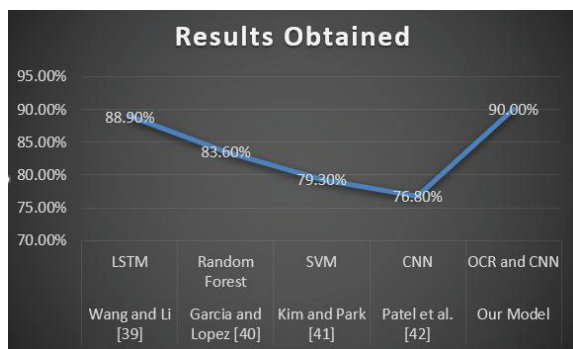


Fig. 9. Comparison graph of proposed model with other ML model in terms of accuracy

V. CONCLUSION AND FUTURE SCOPE

Automated Multi-Page Document Classification and Information Extraction using OCR and CNN is a powerful approach that combines OCR and CNN to tackle the challenges of classification and extraction, potentially saving time and reducing manual effort in various domains such as information retrieval, data analysis, and document management. This method leverages the OCR technology to convert image-based documents into machine-readable text and then utilizes CNN models for efficient and accurate classification and extraction tasks. The results obtained by using Word2vec+CNN, and demonstrate the degrees of accuracy with 90%. Additionally, the effectiveness of the approach can be influenced by factors like the quality of the OCR output, the complexity of the documents, and the diversity of the dataset. The system described in this research paper, shows promising results, achieving high accuracy rates for both tasks. In the future, more advanced OCR and deep learning techniques, to improve accuracy and make the system more robust to variations in image quality. System can be expanded to handle a broader range of document types and formats, such as handwritten documents, which would increase its applicability to various industries and use cases.

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